



Physiology Of Inflammation

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Learning Outcomes:

- ▶ Define inflammation.
- ▶ Describe the stages of inflammation.
- ▶ Explain the signs and symptoms of inflammation.

What is Inflammation?

- ▶ Inflammation is a nonspecific, defensive response of the body to tissue damage.
- ▶ Among the conditions that may produce inflammation are pathogens, abrasions, chemical irritations, distortion or disturbances of cells, and extreme temperature.

signs and symptoms of inflammation

► Four characteristic signs and symptoms of inflammation are:

❑ Redness

❑ Pain

❑ Heat

❑ swelling

► Inflammation can also cause a **loss of function** in the injured area (for example, the inability to detect sensations), depending on the site and extent of the injury.

- Inflammation is an attempt to dispose of microbes, toxins, or foreign material at the site of injury, to prevent their spread to other tissues, and to prepare the site for tissue repair in an attempt to restore tissue homeostasis.

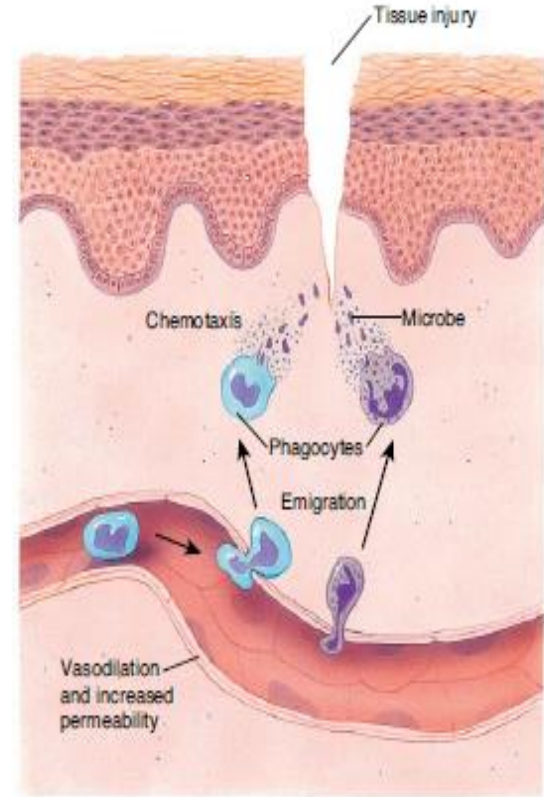


Stages of Inflammatory Response:

- ▶ In each case, the inflammatory response has **Three basic stages**:

1. Vasodilation and increased permeability of blood vessels
 2. Emigration (movement) of phagocytes from the blood into interstitial fluid, and, ultimately
- ↓
3. tissue repair.

🔑 The three stages of inflammation are as follows: (1) vasodilation and increased permeability of blood vessels, (2) phagocyte emigration, and (3) tissue repair.



Phagocytes migrate from blood to site of tissue injury

Vasodilation and Increased Blood Vessel Permeability:

- ▶ Two immediate changes that occur in the blood vessels in a region of tissue injury are:
 1. **vasodilation** which means increase in the diameter of arterioles and allows more blood to flow to the damaged area.
 2. **Increased permeability** which means that substances normally retained in blood are permitted to pass from the blood vessels.
- ▶ The increased blood flow also helps remove microbial toxins and dead cells.
- ▶ Increased permeability permits defensive proteins such as antibodies and clotting factors to enter the injured area from the blood.

Vasodilation and Increased Blood Vessel Permeability:

- ▶ Dilation of arterioles and increased permeability of capillaries produce three of the signs and symptoms of inflammation: **heat, redness (erythema), and swelling (edema)**.
- ▶ **Heat and redness** due to accumulation of fluid. As the local temperature rises slightly, metabolic reactions proceed more rapidly and release additional heat.
- ▶ **Pain** due to injury to neurons and from toxic chemicals released by microbes, or due to increased pressure from edema.
- ▶ fibrinogen is ultimately converted to an insoluble, thick mesh of fibrin threads that localizes and traps invading microbes and blocks their spread.

Substances that contribute to vasodilation, increased permeability, and other aspects of the inflammatory



Histamine

Kinins

Prostaglandins

Leukotrienes

Complement

► **Histamine**: mast cells in connective tissue and basophils and platelets in blood release histamine. Histamine causes vasodilation and increased permeability of blood vessels.

► **Kinins**: Polypeptides formed from kininogens, causes vasodilation and increased permeability and serve as chemotactic agents for phagocytes.



Kinins affect some nerve endings, causing much of the pain associated with inflammation.

- ▶ **Prostaglandins**: intensify the effects of histamine and kinins. PGs also may stimulate the emigration of phagocytes.

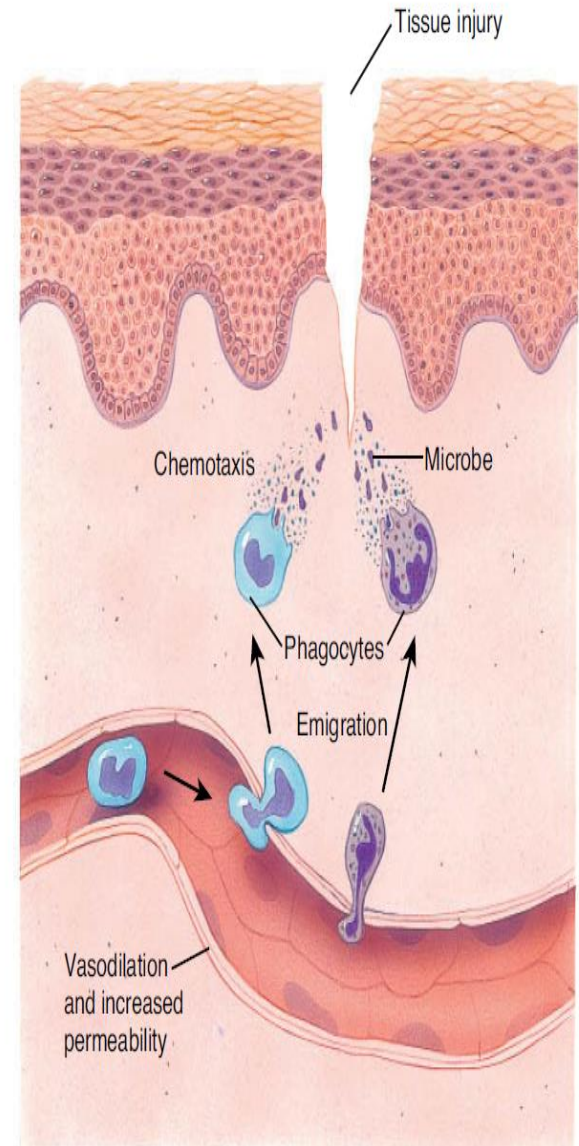


Prostaglandins intensify and prolong the pain associated with inflammation.

- ▶ **Leukotrienes**: they cause increased permeability and attract phagocytes.
- ▶ **Complement**: Stimulate histamine release, attract neutrophils by chemotaxis, and promote phagocytosis.

Emigration of Phagocytes:

- ▶ Within an hour after the inflammatory process starts, neutrophils begin to stick to the inner surface of the endothelium (lining) of blood vessels.
- ▶ Then the neutrophils begin to squeeze through the wall of the blood vessel to reach the damaged area. This process, called **Emigration**, depends on chemotaxis.
- ▶ **Neutrophils** attempt to destroy the invading microbes by **Phagocytosis**.



Emigration of Phagocytes:

- ▶ Neutrophils die, as the inflammatory response continues, **monocytes** follow the neutrophils into the infected area.
- ▶ Once in the tissue, monocytes transform into wandering **macrophages** that add to the phagocytic activity of the fixed macrophages already present.
- ▶ Macrophages are much more potent phagocytes than neutrophils.
- ▶ Eventually, macrophages also die.

What is Pus?

- ▶ **Pus** a pocket of dead phagocytes and damaged tissue forms within a few days after tissue injury and inflammation.
- ▶ Pus formation occurs in most inflammatory responses and usually continues until the infection subsides.
- ▶ At times, pus reaches the surface of the body or drains into an internal cavity and is dispersed.



CLINICAL CONNECTION | Abscesses and Ulcers

- ▶ If pus cannot drain out of an inflamed region, the result is an **abscess**.
- ▶ Abscess is an excessive accumulation of pus in a confined space.
- ▶ When superficial inflamed tissue sloughs off the surface of an organ or tissue, the resulting open sore is called an **ulcer**. People with poor circulation, for instance: diabetics with advanced atherosclerosis, are susceptible to ulcers in their legs.
- ▶ These ulcers, which are called stasis ulcers, develop because of poor oxygen and nutrient supply to tissues that then become very susceptible to infection.



Fever



- ▶ Fever is an abnormally high body temperature that occurs because the hypothalamic thermostat is reset.
- ▶ It commonly occurs during infection and inflammation.
- ▶ Many bacterial toxins elevate body temperature, sometimes by triggering release of fever-causing cytokines such as interleukin-1 from macrophages.
- ▶ Elevated body temperature *intensifies* the effects of interferons, inhibits the growth of some microbes, and speeds up body reactions that aid repair.

THANK
YOU